EC327 Lab 1 Activities

Command Line Basics

Open up the Terminal application. You should see a prompt that looks something like the following:

```
user@hostname:~ $
```

Tasks to demonstrate familiarity with the UNIX command line:

- ☐ Use `echo` to print the string "Hello World!"
- ☐ Navigate to your home directory using `cd`.
- ☐ Make a new directory titled `EC327_Lab_1`. Enter that directory.
- ☐ Create a new file called "hello.txt" using `touch`.
- ☐ List the files in the directory. How would you show all of the files?
- ☐ Open "hello.txt" in your preferred text editor and save a message in it.
- ☐ In the terminal, use `cat` to display the contents of "hello.txt".
- ☐ Change the filename from "hello.txt" to "message.txt" using `mv`.
- ☐ Copy "message.txt" to "goodbye.txt" using `cp`.
- ☐ Open the manual page for `ls` to see what inputs it can take. Use the `man` function.

Once you have shown the course staff these items, you have passed the lab. If you have any trouble, don't hesitate to ask for help.

Bonus 1: Text Processing and Advanced Commands

For this task, we'll be using the file `/usr/share/dict/words`, which exists on most UNIX systems, and contains a list of words in the English language.

`head` and `tail`

- ☐ View the first few words in the UNIX dictionary using `head /usr/share/dict/words`.
- ☐ View the last few words in the UNIX dictionary using `tail /usr/share/dict/words`.

`grep`

- ☐ Open up the manual page for `grep` by typing `man grep`.
- ☐ Search for English words that contain the substring "meow" in `/usr/share/dict/words`.

```
grep meow /usr/share/dict/words
```

Redirects and Sorting

- ☐ Using the >> operator, write the results of the above search to a file called `results.txt`

```
grep meow /usr/share/dict/words >> results.txt
```

- ☐ Open up the manual page for `sort`.
- ☐ Print the contents of "results.txt" in reverse alphabetical order using `sort`

```
sort -r results.txt
```

The W3Schools tutorial for bash contains other useful tricks for using UNIX fluently.

Bonus 2: Git

For this semester, many of the assignments will use the `git` tool for submissions and version control. In Lab 2, we'll go over how to use `git` and the website github.com together.

If you have time before the next lab, read through the linked article: "What is Git?"